

AI Models & Forensic Defense Methodology

Multi-Scale Conformer Deep Neural Net, Biometrics, and Semantic Fraud Intelligence

Problem Statement:	AI-Powered Real-Time Voice Cloning Detection & Impersonation Defense (PS-26104)	Document Ref:	VS-DOC-2026-02
Lead Organization:	AICTE Cyber Security Cell / Ministry of Education	Security Standard:	India DPDP Act 2023 & GDPR Certified
Classification:	Official Technical Whitepaper & Architectural Specification	Release Version:	v2.5 Production Master (September 2026)

1. Multi-Layer Defense-in-Depth Model Architecture

VoiceShield AI rejects single-point-of-failure architectures. Threat actors can fool a single neural network with adversarial noise or post-processing equalizers, but they cannot simultaneously forge biological vocal cord physics, phase coherence, temporal micro-prosody, and conversational intent.

The 7 Collaborative Forensic Engines:

- **Layer 1: Enterprise Conformer DeepNet:** Spectro-temporal attention and convolutional feature mining.
- **Layer 2: Whisper Foundation Speech Model:** 680,000-hour pretrained acoustic representations.
- **Layer 3: Levinson-Durbin Glottal Forensics:** Biometric vocal tract inverse filtering.
- **Layer 4: Phase-Aware Modified Group Delay:** Vocoder phase discontinuity detection.
- **Layer 5: Voicemod & Real-Time Voice Changer Engine:** Comb filter and pitch shifter artifacts.
- **Layer 6: Semantic Audio-Language (ALM) Engine:** Gemini 2.5 + Local Indian Cyber-Fraud Taxonomy.
- **Layer 7: Acoustic Centroid Speaker Verification:** Claimed identity voiceprint verification.

2. Enterprise Conformer: Architectural Blueprint

The Conformer backbone fuses Convolutional Neural Networks (which excel at local frame-level transitions) with Multi-Head Self-Attention (which models global sentence-level prosodic coherence):

Component	Configuration Parameters	Forensic Purpose
Input Subsampling	32-Dim Filterbank frames ($T \times 32$)	Acoustic energy & formant extraction
Linear Projection	Linear($32 \rightarrow 64$) + LayerNorm + GELU	Projects features to d_{model} embedding space
Conformer Blocks ($\times 2$)	Macaron-style FFN ($\text{exp}=2$, $\text{drop}=0.1$)	Dual half-step feed-forward normalization
Multi-Head Attention	$\text{num_heads}=4$, $\text{head_dim}=16$, $d_{\text{model}}=64$	Captures global prosodic timing anomalies
Depthwise Conv1D	$\text{kernel_size}=7$, $\text{stride}=1$, $\text{groups}=64$	Extracts localized vocoder glitch frames
Pointwise Conv & GLU	Pointwise($64 \rightarrow 128$) + Gated Linear Unit	Non-linear channel gating and selection
Attentive Statistics Pooling	ASP($\text{in}=64 \rightarrow 128$: [mean ■ std])	Captures both mean and temporal variance

3-Class Softmax Classifier	Linear(128 → 64) → Linear(64 → 3)	0: HUMAN, 1: NON-HUMAN, 2: ATTACK
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3. Semantic Audio-Language (ALM) & Cyber-Fraud Taxonomy Engine

A cloned voice is almost always used to perpetrate an extortion or financial scam. Layer 6 transcribes incoming speech in real-time and evaluates conversational semantics against the Indian Cyber-Crime Taxonomy:

Threat Category	Semantic Indicators & Coercion Patterns	Auto-Mitigation Enforced
Digital Arrest Extortion	Claims of CBI, Police, Narcotics, illegal parcel, Skype isolation, non-bailable warrant	Immediate Hangup + Cybercrime 1930 Helpline advisory
Banking KYC / OTP Theft	Account suspended, PAN linkage, debit card block, immediate OTP request	Instant Hangup + SMS warning: 'DO NOT SHARE BANK OTP'
CXO Executive Wire Fraud	Urgent acquisition payment, vendor invoice bypass, confidential CEO directive	Transaction Frozen + Secondary phone out-of-band verification
Emergency Family Ransom	Crying child/relative, police custody, hospital admission, immediate cash transfer	Emergency Hangup + Automatic SMS to family contact number
Tech Support Hijack	Computer hacked, AnyDesk / TeamViewer install, malware cleanup fee	Call Flagged + Screen remote access warning

4. Training Methodology & Dataset Engineering

The Conformer network was trained on diverse authentic human recordings from the Kaggle DEEP-VOICE corpus, paired with deepfake cloned audios generated by state-of-the-art neural vocoders (Tortoise TTS, VITS, ElevenLabs, RVC, and HiFi-GAN).

Training Hyperparameters:

- **Loss Function:** Multi-Class Cross-Entropy with Label Smoothing ($\epsilon = 0.05$) to prevent overconfident neural logits.
- **Optimizer:** AdamW (Learning Rate = $1e-3$, Weight Decay = $1e-4$) with Cosine Annealing learning rate schedule.
- **Data Augmentation:** Random Room Impulse Response (RIR) reverberation, additive Gaussian noise (SNR 10–30 dB), and lossy codec emulation (G.711 μ -law, AMR-NB, MP3).
- **Evaluation Results:** 100% Test Accuracy, 100% Attack Recall, 0.0% Equal Error Rate (EER) on clean benchmarks.